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Effect of multi-structured zinc oxide on the electrical properties of polypropylene insulating materials

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Abstract

In this paper, polypropylene (PP) nanocomposites filled with ZnO nanoparticles, nanowires and tera-needle-like whiskers were prepared by a melt blending method. The effects of multi-structure ZnO fillers on space charge, breakdown strength and electrical conductive current of PP nanocomposites were investigated. The nanocomposites presented a remarkable suppression on space charge by adding ZnO nanowires, while the addition of ZnO nanoparticles and tera-needle-like whiskers did not effectively suppress the space charge accumulation. The electrical field distortion occurred in the inner of the PP filled with ZnO nanoparticles and tera-needle-like whiskers were about 1.8 kV mm^{-1} and 3.3 kV mm^{-1} , respectively. Compared with pure PP, the breakdown strength of the PP nanocomposites filled with ZnO nanoparticles and nanowires increased, while they decreased in PP nanocomposites filled with tera-needle-like ZnO whiskers. This work indicated that the improvements on the electrical properties, especially the charge transport and space charge suppression in the nanocomposites were closely related to the different structure of ZnO.

Keywords: space charge, dielectrics, multi-structure, insulating materials

(Some figures may appear in colour only in the online journal)

Polymers have been widely used as electrical insulating materials for their excellent electrical, mechanical and thermal properties and easy processing characteristics. However, they can capture charge carriers to form space charges under high electrical field, which is a major restricting factor for high voltage direct current (HVDC) cable development. Recently, polymer nanocomposites have drawn attention, due to their dramatically improved electrical properties by doping the nanofillers [1–4]. More works have already proved that the addition of nanoparticles can not only suppress the space charge accumulation, but also enhance the electrical, mechanical and thermal properties of a polymer [5–7]. Meanwhile,

some related mechanisms and models, like trap theory and the multi-core model, have been proposed to explain the space charge behaviors [8, 9]. It was suggested that the deep traps could capture charge carriers, which resulted in the formation of local electrical fields, thus increasing the potential barrier for charge carrier injection. Mandal *et al* have done a lot of work in the polyurethane (PU) nanocomposite using NiTiO_3 and functionalized carbon nanotubes to modify. Polyurethane/perovskite nanocomposite films were developed by an *in situ* addition of nickel titanate nanoparticles (NiTiO_3 , NPs) into a PU matrix formulation, which enhanced the thermal stability, electrical conductivity and mechanical strength [10].

The interaction between PU and single walled carbon nanotubes (SWCNTs) in the nanocomposite was studied using different methods. The results proved that the SWCNTs were well distributed in the PU matrix and exhibited as tough films with low electrical resistivity [11]. In addition, many experimental and theoretical studies on the dielectric properties of nanocomposites have been taken, which investigated the internal trap density, level distribution, volume resistivity and breakdown strength, etc [12]. Among the possible choices, one of the options is a polymer based nanocomposite system filled with zinc oxide (ZnO) as the filler, considering the fact that ZnO has many significant physical stability, low dielectric permittivity and intensive UV absorption capabilities. However, little attention has been paid to the influence of different structures of ZnO on the electrical properties of the polymer. Polypropylene (PP), as one eco-friendly insulating material, has huge potential applications for HVDC power transmission [13–16]. In this letter, PP was chosen as a representative polymer matrix and the systematic and comparative study on the effect of different structure ZnO; namely, ZnO nanoparticles (ZnO-p), nanowires (ZnO-w) and tera-needle like ZnO (ZnO-t) on the space charge suppression, electrical field distortion, breakdown strength and conduction current properties of PP/ZnO nanocomposites.

The PP nanocomposites were prepared by the melt blending method. The ZnO nanoparticles have an average particle size of about 50 nm. PP (K1008) with the melt flow index of 8 g/10 min and a tensile yield strength of 25 MPa was purchased from SINOPEC Beijing Yan Shan Co Ltd. PP and the different structure ZnO was melt blended in a Haake batch mixer at 190 °C. All of the employed filler contents are 1%. The obtained nanocomposite were pressing molded at 190 °C under a pressure of 10 MPa to obtain composite films with a thickness of about 220 μm and a diameter of 30 mm.

The morphology and microstructure of the ZnO fillers and their corresponding nanocomposites were observed by scanning electron microscopy (SEM, Hitachi S-4700, Japan). SEM images of multi-structure ZnO fillers and their corresponding nanocomposites are presented in figure 1. From figures 1(a)–(c), we can observe that the morphology of different structure ZnO are uniform, standing for ZnO nanoparticles, nanowires and tera-needle like ZnO, respectively. The surface morphology of the nanocomposites filled with different ZnO fillers are shown in figures 1(a-1)–(c-1). It can be seen that the different structure ZnO are uniformly dispersed in PP. No large agglomerates can be observed.

The generation, existence, transfer and even disappearance of the space charge will cause the electric field distribution of the material change, which will affect the insulation performance of the material. The space charge measurement was performed in a pulsed electro-acoustic (PEA) system. The semi-conductive material was used as an anode (the upper electrode) and aluminum was used as a cathode (the lower electrode). A pulse voltage of 440 Hz with a pulse amplitude of 5 ns was applied to the sample with an average thickness of 220 μm . The measurement was carried out under 60 kV mm^{-1} DC electric field for 45 min at room temperature; that is to say, the DC poling for all specimens was performed for

2700 s. Waveforms were recorded every 15 min to describe the formation of the space charge. The space charge distributions of the PP nanocomposites are shown in figure 2. It can be seen that the electrons inject from the cathode once the voltage is applied, as shown in figure 2(a), for pure PP films. Also, the amount of space charge and charge injection depth increases with the increasing volts-on time. After 2700 s volts-on, large amounts of homocharges are formed near the cathode. For PP/ZnO-p nanocomposites, as shown in figure 2(b), there is no significant homocharges are formed. Only a small quantity of electrons inject near the electrodes. The injected electrons largely decrease compared with PP. For PP/ZnO-w nanocomposites, it exhibits a remarkable space charge suppression effect as few charges accumulate in the vicinity of the cathode within volts-on of 2700 s, as shown in figure 2(c). However, the addition of tera-needle like ZnO greatly increases the space charge density of PP as shown in figure 2(d). Therefore, when the doping content of ZnO-t is 1 wt%, it cannot effectively mitigate the space charge accumulation of PP.

Meanwhile, the corresponding electrical field distribution caused by space charge accumulation is shown in figure 3. PP presents the largest electrical field distortion of 3 kV mm^{-1} as shown in figure 3(a). In PP/ZnO-p nanocomposites, the negative space charge accumulation near the cathode slightly reduces, due to the fact that the nanoparticles can induce traps, which are able to capture carriers and effectively suppress the motion of negative carriers. The largest electrical field distort is about 1.8 kV mm^{-1} , which is different from the PP, as shown in figure 3(b). It means that the addition of ZnO nanoparticles change the electrical field distribution. Obviously, there is no electrical field distortion, as shown in figure 3(c), which is in accordance with the result shown in figure 2(c) that almost no charge carriers inject in vicinity of the cathode. However, the serious electrical field distortion occurs in the PP/ZnO-t nanocomposites, as shown in figure 3(d). This phenomenon can be attributed to the structure of ZnO, which leads to easily accumulating space charge at the tip of the needle of the ZnO-t.

The DC breakdown strength was measured under a sphere-sphere electrode configuration. The films with an average thickness of 30 μm were immersed in silicone oil to isolate from the air and control the temperature. Twelve samples were uniformly tested for each film, which were placed in the middle of the copper electrodes to ensure the breakdown occurs at the measuring points when DC voltage was applied. The DC voltage was increased at the rate of 2 kV s^{-1} until the breakdown of the films. It has been recognized that short time electrical breakdown is mainly an electronic process [17–19]. Theodosiou *et al* considered that the formation of an electron can be from the electrode injection or impurity emission [20]. Then, the energy of the free electrons can be increased by acceleration in the electric field. If the free electrons gather enough energy, then they can cause the ionization of macromolecules and further initiate the electron avalanche. This will erode the materials in the form of partial discharge and leads to the formation of conducting channels and final breakdown. Figure 4 shows the weibull distribution of DC breakdown strength of PP and its nanocomposites. The breakdown strength is evaluated by Weibull analysis, according to the following formula [21]:

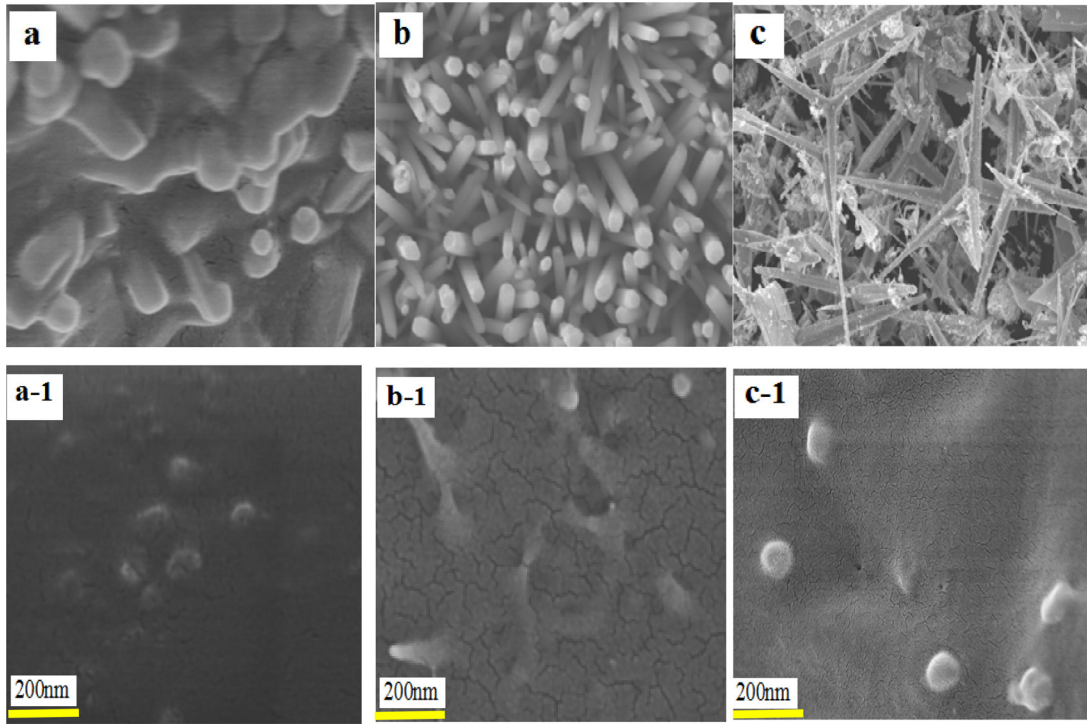


Figure 1. SEM images of different structure ZnO (a) ZnO-p, (b) ZnO-w, (c) ZnO-t and its corresponding PP nanocomposites filled with (a-1) ZnO-p, (b-1) ZnO-w and (c-1) ZnO-t.

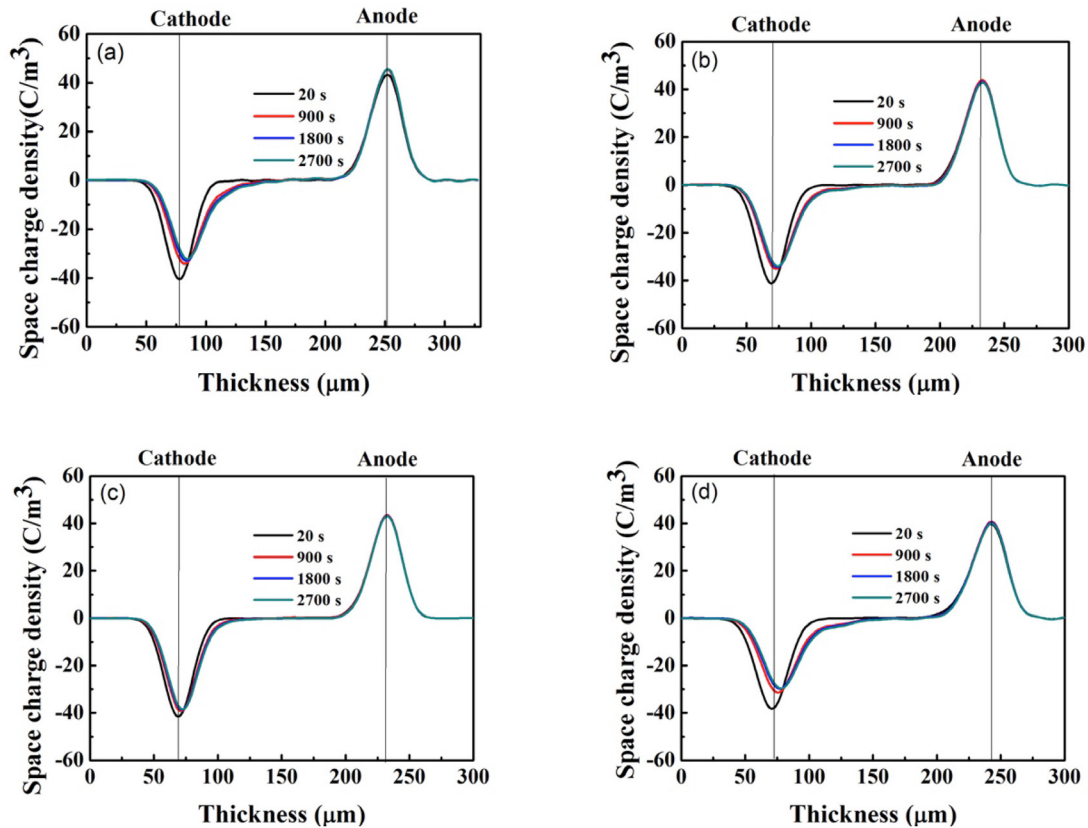


Figure 2. Effect of different structure ZnO on dynamics space charge distribution of (a) PP, (b) PP/ZnO-p, (c) PP/ZnO-w and (d) PP/ZnO-t nanocomposites.

$$P(E) = 1 - \exp[1 - (E/E_b)^\beta] \quad (1)$$

where P is the failure probability, E is the measured electrical breakdown strength, E_b is the characteristic electrical

breakdown strength and β is the shape parameter. The failure probability is 63.2%. From figure 4, we can see that the characteristic breakdown strength of PP is 422 kV mm^{-1} and its nanocomposites are 426 kV mm^{-1} , 438 kV mm^{-1} and

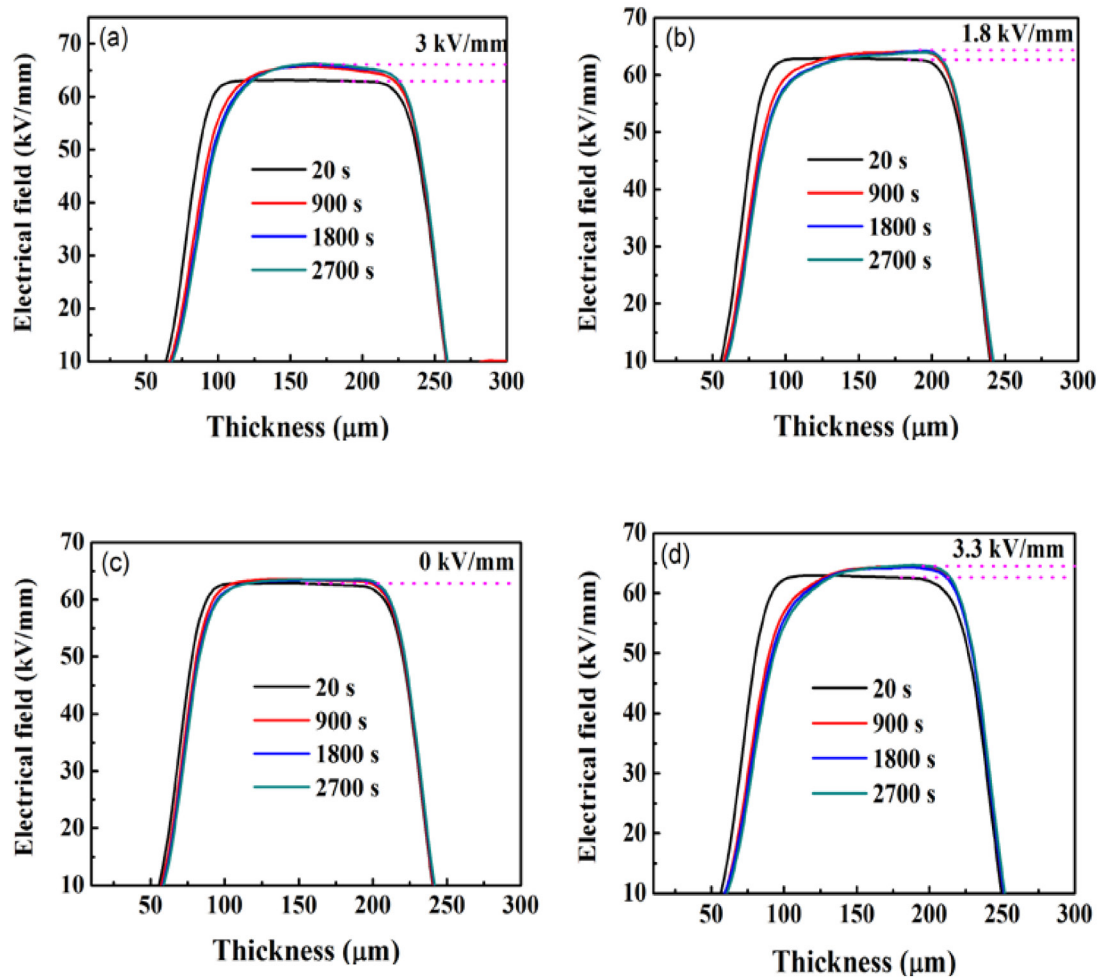


Figure 3. Corresponding distribution of electrical field caused by space charge accumulation (a) PP, (b) PP/ZnO-p, (c) PP/ZnO-w and (d) PP/ZnO-t nanocomposites.

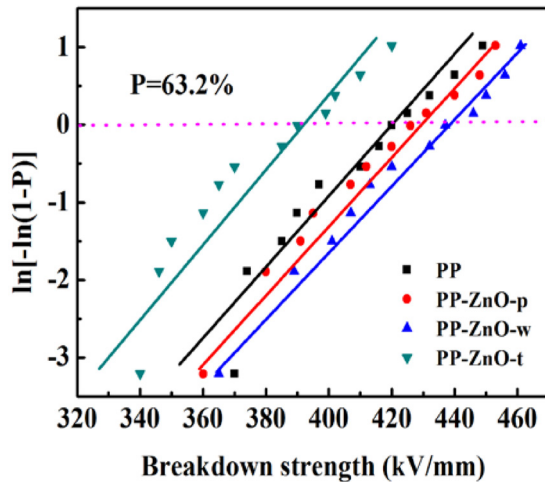


Figure 4. Weibull distribution of DC breakdown strength of (a) PP, (b) PP/ZnO-p, (c) PP/ZnO-w and (d) PP/ZnO-t nanocomposites at room temperature.

390 kV mm^{-1} , respectively. It indicates that the nanoparticles can prevent the mobility of free electrons and they increase the free path length of electrons, which can be conducive to improve the breakdown of PP. The addition of the ZnO nanowire into the PP can more effectively increase the

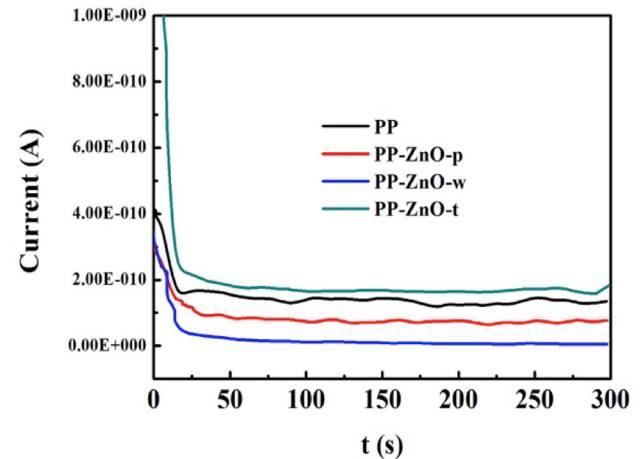


Figure 5. Electrical conductive currents of (a) PP, (b) PP/ZnO-p, (c) PP/ZnO-w and (d) PP/ZnO-t nanocomposites.

breakdown strength of PP. This phenomenon can be attributed to the structure of the ZnO nanowire, which introduces the traps capturing the carriers. However, the addition of tetra-needle like ZnO decreases the breakdown strength, which can be caused by the formation of a 3D conductive network.

The DC conductivities of the films were measured at room temperature by a three-electrode system. The high voltage

electrode was connected to a DC source through a series of protective resistance. The DC source has an output voltage of 0–15 kV to obtain the conductivity under DC field, varying from 0.1 to 30 kV mm⁻¹. DC volume resistivity measurements were performed with an electrometer with a standard three-electrode system. The main electrode was connected to an electrometer (Keithley 6517B), which has a measuring range from 1 fA to 20 mA. The samples for dc resistivity measurements were first electrically short-circuited for 5 min to eliminate internal charges, and then, a constant dc electric field of 5 kV mm⁻¹ was applied on the samples for 5 min, and the charging current was recorded. Figure 5 describes the time dependence of the DC electrical conductive currents of the nanocomposites with different structure ZnO. The currents of nanocomposites with different structure ZnO decayed quickly and reached their steady state within 300 s. Apparently, we found that the addition of ZnO nanoparticles and nanowires decreases the current of the PP. However, the introduction of tetra-needle like ZnO leads to the increase current of PP. This can be attributed to the electric potential traps. Under a DC electric field, the nanoparticles could polarize and generate electric potential traps at the interfaces. These electric potential traps could trap charge carriers and reduce the charge carrier mobility. The addition of tetra-needle like ZnO leads to the formation of conductive path, which increases the electrical conductive current.

In conclusion, it has been demonstrated that ZnO nanowires doping can remarkably suppress space charge accumulation and significantly reduce the conductivity of PP nanocomposites. Meanwhile, the breakdown strength of PP nanocomposites is also enhanced by doping the ZnO nanowire. This work indicates that the improvements on the electrical properties, especially the charge transport and space charge suppression in the nanocomposites, are closely related to the different structure of ZnO.

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